

12.788

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

$$\mathbf{L} = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & 0 \\ -1 & 0 & -1 & 1 \end{pmatrix}$$

Which of the following are TRUE?

a) The matrix $\mathbf{L}$ is row equivalent to $\begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & 0 \\ -1 & 0 & -1 & 1 \end{pmatrix}$

b) The linear system $\mathbf{Lx} = \mathbf{b}$ has a solution for all $\mathbf{b}$

c) For $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, the system $\mathbf{Lx} = \mathbf{b}$ has a solution

d) Rank of the matrix $\mathbf{L}$ is 3

Solution:

Given:

$$\mathbf{L} = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & 0 \\ -1 & 0 & -1 & 1 \end{pmatrix} \quad (1)$$

Row reduction (Option a):

$$R_1 \rightarrow R_1 + R_2 + R_3 + R_4 \implies \mathbf{L} \sim \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & 0 \\ -1 & 0 & -1 & 1 \end{pmatrix} \quad (2)$$

Rank of $\mathbf{L}$ (Option d):

$$\text{One row is dependent, remaining 3 independent} \implies \text{rank}(\mathbf{L}) = 3 \quad (3)$$

System $\mathbf{Lx} = \mathbf{b}$ for all $\mathbf{b}$ (Option b):

$$\text{Rank}(\mathbf{L}) = 3 < 4 \implies \text{No solution} \quad (4)$$

Dependence property (Option c):

$$R_1 + R_2 + R_3 + R_4 = \mathbf{0} \quad (5)$$

$$\implies (R_1 + R_2 + R_3 + R_4)\mathbf{x} = 0 \quad (6)$$

$$\implies R_1\mathbf{x} + R_2\mathbf{x} + R_3\mathbf{x} + R_4\mathbf{x} = b_1 + b_2 + b_3 + b_4 = 0 \quad (7)$$

$$\text{For } \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad b_1 + b_2 + b_3 + b_4 = 4 \neq 0 \implies \text{no solution} \quad (8)$$